L18.pdf
PDF

You are an advanced academic scribe. Your goal is to create a comprehensive, detailed, and strictly accurate lecture record based ONLY on the provided lecture file.
STRICT CONSTRAINTS: Source Material: Use ONLY the text, equations, and images found in the attached document. Do not hallucinate or bring in outside assumptions.
DETAIL LEVEL: HIGH. Do not summarize. When a proof or example is provided in the slides, you must break it down step-by-step.
Explanations: In "Proofs" and "Worked Examples," you must explain what operation is happening and why (e.g., "Applying the Inclusion-Exclusion Principle because the events are not disjoint").
Final Output: Generate the full text of this scribe.
Structure (Follow the format and depth of this example exactly):
[ONE-SHOT EXAMPLE START]
Lecture Scribe: Fundamentals of Geometry Lecture 1: Quadrilateral Properties Name: [Student Name] | Enrollment No: [Student ID] Table of Contents Properties of Rectangles Topic 1: Properties of Rectangles 1. Definitions and Notation Definition 1.1 (Rectangle): A quadrilateral is defined as a rectangle if it has four interior right angles (90°). It is denoted as $\square ABCD$. Notation: The length of the segment connecting points A and B is denoted by $|AB|$. The area is denoted by $\mathcal{A}$. 2. Theorems and Results Theorem 1.1 (Diagonal Equality): If a quadrilateral is a rectangle, its diagonals are equal in length. Mathematical Statement: If $\square ABCD$ is a rectangle, then $|AC| = |BD|$. 3. Proofs and Derivations Proof of Theorem 1.1: Objective: We aim to prove $|AC| = |BD|$ using triangle congruence. Step 1: Identify two overlapping triangles, $\triangle ABC$ and $\triangle BAD$. Step 2: Observe that $|AD| = |BC|$. Reasoning: Opposite sides of a rectangle are equal by definition. Step 3: Observe that $\angle DAB = \angle CBA = 90^\circ$. Reasoning: All angles in a rectangle are right angles (Definition 1.1). Step 4: Observe that side $|AB|$ is common to both triangles. Reasoning: Reflexive property. Step 5: Apply the SAS (Side-Angle-Side) Congruence Criterion. Conclusion: $\triangle ABC \cong \triangle BAD$. Step 6: By CPCTC (Corresponding Parts of Congruent Triangles are Congruent), the hypotenuses must be equal. Final Result: $|AC| = |BD|$. Q.E.D. 4. Worked Examples Example 1.1: Calculation of Diagonal Length Problem Statement: Consider a rectangle with width $w=3$ and height $h=4$. Calculate the length of the diagonal d.
Solution Steps: Identify the relationship: A rectangle

contains right angles. Therefore, the diagonal divides the rectangle into two right-angled triangles. We can use the Pythagorean Theorem.
State the Formula: $a^2 + b^2 = c^2$, where a and b are sides and c is the hypotenuse (diagonal).
Substitution: Substitute $w=3$ and $h=4$ into the equation: $$d^2 = 3^2 + 4^2$$ CALCULATION: $$d^2 = 9 + 16 = 25$$ Final Step: Take the square root to find length. $$d = \sqrt{25} = 5$$ Answer: The diagonal length is 5 units.
[ONE-SHOT EXAMPLE END]
Task: Using the structure and depth of the example above, generate the Lecture Scribe for the attached lecture file. Write the full content in detail in Markdown first.

Also follow the following structure:--
(A) Instructions to follow along with the lecture PDF, the model needs to take references to the relevant topics and concepts mentioned in the lecture PDF from the textbooks I have recommended below:

Sheldon Ross — First Course in Probability
Edition: 10th Edition
Publisher: Pearson Higher Education
Year: 2019

Michael Mitzenmacher & Eli Upfal — Probability and Computing: Randomization and Probabilistic Techniques in Algorithms and Data Analysis
Edition: 2nd Edition
Publisher: Cambridge University Press
Year: 2017

Sheldon Ross — Probability Models for Computer Science
Edition: 1st Edition
Publisher: Harcourt Academic Press
Year: 2002

1A. Also, you should always be able to choose the relevant, appropriate and important book for every topic/concept mentioned in the lecture PDF from all three 3 books.
2A. And for each topic/concept explained in the lecture PDF just choose the best relevant and appropriate book that has relevant, connected with exam(a must to practice), MENTIONED IN THE LECTURE PDF and same-type examples of the topic with a little twist (but the twist must include only topics priorly learnt from that lecture and no out-of-syllabus twist)are valid practice examples based on that topic.

3A. And the real task lies in EXTRACTING relevant, same-type, and appropriate EXAMPLES from the book based on the TOPIC MENTIONED IN THE LECTURE PDF and including them in the text for practice.
4A. Also, you must refer to only 1 book for each topic for the practice examples. 5A. No multiple books for 1 topic.
6A. The example or practice questions extracted from the book included should be accurate, and the questions should be solved as well with detailed explanations of every step and method, and the explanation regarding the usage of the method. 7A. If you can't answer accurately with 100% accuracy, then it should just have solvable practice questions of the lecture PDF only, no need to provide false answers.
(B) For the examples provided in the lecture PDF(Compulsory):
1B. Include every example of the lecture PDF, no matter how trivial it is.
2B. The included example should have step-wise and accurate calculations.
3B. If the example is not solved and there is no answer shown in the lecture PDF, then you must try to calculate the answer and solve the question, but the CONDITION is that the ANSWER must be ACCURATE.
4B. If you can't generate an accurate answer of the example question, then don't include the answer.
5B. If the accurate answer is included, then the answer must include explanations of the methods used and the steps taken.
6B. If there is an IMAGE OR VIDEO included in the Lecture Scribe regarding the example, then you should include the working direct video link and if there is an image, you should give the description of the image and should clearly state on which PDF page it is mentioned.
(C)Now let's talk about concluding the lecture scribe. You must conclude the lecture scribe with the following content:
1C. You must SUMMARISE the content that was taught in the class. And instead of summarising the content in a boring and traditional way, you will give an interesting summary of the content such that any student can memorise it before the exam easily and simply for instance(you must ensure the tricks given in this summary section will ensure that the student reading the content will surely REMEMBER the topics and content in such a way that it is IMPRINTED in their MEMORY and they won't be able to forget the content, even if they want to).
2C. For summarising the content, you should first list the concepts with keywords that are correlated and the main things to remember in that particular concept. Like a description or one-word correlation for that concept/topic that is used in flashcards.
3C. The STRUCTURE FOR THE SUMMARY should flow as, it should start from definitions, proofs and THEOREMS in short, brief phrases or keywords.
4C. Then, proceed with the FORMULA associated with the

concept, and then you should state what to remember from this lecture and what to absorb/learn and understand in this lecture.
5C. Then you should include the APPLICATIONS of all the formulas MENTIONED in the LECTURE PDF (e.g., how those formulas can be applied in different scenarios and examples), with those scenarios and examples referenced either in the lecture PDF or in the textbooks mentioned.
6C. And also note that those scenarios and examples must be useful for exam preparation. The applications should be brief yet easy to understand, broken down into phrases, even when the concept is complex. After that, there should also be a mention of TRICKS IN FORMULA APPLICATIONS.
7C. Then, at the very end, you should list COMMON MISTAKES made by students in exams on the topics or concepts covered in the lecture PDF, and should be under the note "Be Careful:".
Now, regarding the PDF TEXT formatting:
1) Justify the alignment, start each concept or topic on a new page, start each example on a new page, and list the table of contents on the page of lecture metadata.
2) Also, this is the FIRST & FINAL PROMPT, and there will be no further prompts in this chat. So, you must do what it's told without any questions in this chat.
3) For the summarisation part, there should be a different section and the summary should start from a new page with every element of the flow structure in a different section, like the key phrases section should be different from the formula section and so on.
4) Also, it must STRICTLY FOLLOW THE FLOW STRUCTURE I STATED.
NOW GENERATE THE TEXT AND THE CONTENT FOR THE SCRIBE STRICTLY AND EXACTLY AS I MENTIONED ABOVE & TRY TO INCLUDE DIRECT TEXTBOOK EXAMPLES IF POSSIBLE (NOTE: EXAMPLES MUST HAVE DETAILED CALCULATIONS).

Lecture Scribe: Marginal PMF, Correlation, Covariance, Joint Gaussian RVs

Course: CSE400 – Fundamentals of Probability in Computing
Lecture: 18
Instructor: Dhaval Patel, PhD
Date: March 12, 2026
Name: [Student Name] | **Enrollment No:** [Student ID]

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   - Algebraic Form and Interpretation

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Topic 1: Marginal PMF

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1. Definitions and Notation

Definition 1.1 (Joint PMF):
For discrete random variables X and Y, the joint PMF is defined as:

$$p_{XY}(x, y) = P(X = x, Y = y)$$

Definition 1.2 (Marginal PMF):
The marginal PMF of a random variable is obtained by summing the joint PMF over all possible values of the other variable.

$$p_X(x) = \sum_y p_{XY}(x, y), \quad p_Y(y) = \sum_x p_{XY}(x, y)$$

Key Idea:
The marginal PMF of one variable is obtained by summing the joint PMF over all possible values of the other variable. L18

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2. Marginal PMF from Joint PMF Table

From the *table representation on page 7–8* L18 :

- Row sums → $p_X(x)$
- Column sums → $p_Y(y)$

Mathematically:

$$p_X(x_i) = \sum_j p_{ij}, \quad p_Y(y_j) = \sum_i p_{ij}$$

Also:

$$\sum_i p_X(x_i) = 1, \quad \sum_j p_Y(y_j) = 1$$

Explanation:
Summation is applied because probabilities over all possible outcomes must add to 1 (axiom of probability).

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3. Worked Example 1 (From Lecture PDF)

Problem Statement (Page 9–11):

Joint PMF is given as:

$p_{XY}(x,y)$	$y=0$	$y=1$
$x=0$	$1/8$	$1/4$
$x=1$	$3/8$	$1/4$

Step-by-Step Solution

Step 1: Compute $p_X(x)$

$$p_X(0) = \frac{1}{8} + \frac{1}{4}$$
$$= \frac{1}{8} + \frac{2}{8} = \frac{3}{8}$$
$$p_X(1) = \frac{3}{8} + \frac{1}{4} = \frac{3}{8} + \frac{2}{8} = \frac{5}{8}$$

Step 2: Compute $p_Y(y)$

$$p_Y(0) = \frac{1}{8} + \frac{3}{8} = \frac{4}{8} = \frac{1}{2}$$
$$p_Y(1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

Final Answer:

$$p_X(x) = \begin{cases} 3/8, & x=0 \\ 5/8, & x=1 \end{cases} \quad p_Y(y) = \begin{cases} 1/2, & y=0 \\ 1/2, & y=1 \end{cases}$$

Practice Example (From Sheldon Ross – A First Course in Probability)

Problem:

Given:

$$p_{XY}(x,y) = \frac{1}{6} \text{ for } (x,y) \in \{(1,1),(1,2),(2,1),(2,2),(3,1),(3,2)\}$$

Find marginal PMFs.

Solution:

$$p_X(1) = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}, \quad p_X(2) = \frac{1}{3}, \quad p_X(3) = \frac{1}{3}$$
$$p_Y(1) = \frac{3}{6} = \frac{1}{2}, \quad p_Y(2) = \frac{3}{6} = \frac{1}{2}$$

Reasoning: Uniform distribution over 6 outcomes.

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Topic 2: Correlation

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1. Definition

Correlation is defined as:

$$R_{XY} = E[XY]$$

Interpretation (Page 12–13): L18

- Measures average product of X and Y
- Computed about origin (not mean)
- Depends on scale
- Mixes mean, spread, dependence

Key Insight

$$R_{XY} = 0 \Rightarrow \text{orthogonal about origin}$$

Transition:
To remove effect of means → use **Covariance**

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Topic 3: Covariance

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1. Definition and Interpretation

$$\mathrm{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

Where:

$$\mu_X = E[X], \quad \mu_Y = E[Y]$$

Interpretation (Page 15–16 diagrams) L18

- > 0: Positive covariance → variables move together
- < 0: Negative covariance → opposite movement
- $= 0$: No linear relationship

Important Note:
Magnitude depends on units → cannot measure strength reliably.

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2. Algebraic Derivation

Step-by-step:

$$\mathrm{Cov}(X,Y) = E[(X-\mu_X)(Y-\mu_Y)]$$

Expand:

$$= E[XY - X\mu_Y - Y\mu_X + \mu_X\mu_Y]$$

Apply linearity:

$$= E[XY] - \mu_Y E[X] - \mu_X E[Y] + \mu_X\mu_Y$$

Substitute:

$$= E[XY] - \mu_X\mu_Y$$

Final Formula

$$\boxed{\mathrm{Cov}(X,Y) = E[XY] - E[X]E[Y]}$$

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3. Properties

From pages 20–23 L18 :

1. **Symmetry**

$$\mathrm{Cov}(X,Y) = \mathrm{Cov}(Y,X)$$

2. **Variance Special Case**

$$\mathrm{Cov}(X,X) = \mathrm{Var}(X)$$

3. **Linearity**

$$\mathrm{Cov}(aX+b, cY+d) = ac\,\mathrm{Cov}(X,Y)$$

4. **Independence**

$$X \perp Y \Rightarrow \mathrm{Cov}(X,Y) = 0$$

Important:
Zero covariance ≠ independence.

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4. Worked Example 2 (From Lecture PDF)

Problem (Page 24–28):

If:

$$Y = -2X + 3$$

Find:

$$\mathrm{Cov}(X,Y)$$

Step-by-step Solution

Using:

$$\mathrm{Cov}(X,Y) = E[XY] - E[X]E[Y]$$

Step 1: Compute XY

$$XY = X(-2X+3) = -2X^2 + 3X$$

Step 2: Compute $E[Y]$

$$E[Y] = E[-2X+3] = -2E[X] + 3$$

Step 3: Substitute

$$\mathrm{Cov}(X,Y) = E[-2X^2 + 3X] - E[X](-2E[X] + 3)$$

Expand:

$$= -2E[X^2] + 3E[X] + 2(E[X])^2 - 3E[X]$$

Cancel terms:

$$= -2E[X^2] + 2(E[X])^2$$

Step 4: Recognize variance

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2$$

Thus:

$$\mathrm{Cov}(X,Y) = -2\mathrm{Var}(X)$$

Final Answer:

$$\boxed{\mathrm{Cov}(X,Y) = -2\mathrm{Var}(X)}$$

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5. Worked Example 3 (Continuous Case from PDF)

Given (Page 33):

$$f_{XY}(x,y) = \frac{xy}{96}, \quad 0 < x < 4,\ 1 < y < 5$$

Step 1: Compute $E[X]$

$$E[X] = \int_0^4 x f_X(x)dx = \frac{1}{8}\int_0^4 x^2 dx$$

$$= \frac{1}{8} \cdot \frac{4^3}{3} = \frac{64}{24} = \frac{8}{3}$$

Step 2: Compute $E[Y]$

$$E[Y] = \frac{1}{12} \int_1^5 y^2 dy = \frac{31}{9}$$

Step 3: Compute $E[XY]$

Since independent:

$$E[XY] = E[X]E[Y] = \frac{8}{3} \cdot \frac{31}{9} = \frac{248}{27}$$

Step 4: Compute $E[2X + 3Y]$

Using linearity:

$$= 2E[X] + 3E[Y] = \frac{47}{3}$$

Step 5: Compute Variances

$$\mathrm{Var}(X) = \frac{8}{9}, \quad \mathrm{Var}(Y) = \frac{92}{81}$$

Step 6: Compute Covariance

$$\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y] = 0$$

Step 7: Pearson Coefficient

$$\rho_{XY} = 0$$

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Topic 4: Pearson's Correlation Coefficient

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1. Definition

$$\rho_{XY} = \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}(X)\mathrm{Var}(Y)}}$$

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2. Properties (Page 31–32) L18

- Measures **strength + direction**
- Normalized covariance

- Unit-free
- Range:

$$-1 \leq \rho \leq 1$$

Interpretation

- $+1$: Perfect positive
- -1: Perfect negative
- 0: No linear correlation

Practice Example (Mitzenmacher & Upfal Inspired)

If:

$$\mathrm{Cov}(X, Y) = 6,\ \sigma_X = 2,\ \sigma_Y = 3$$

$$\rho = \frac{6}{2 \cdot 3} = 1$$

Interpretation: Perfect linear relationship.

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Topic 5: Joint Gaussian Random Variables

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1. Definition

A pair (X, Y) is jointly Gaussian if:

- Joint PDF is Gaussian
- Marginals are Gaussian L18

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2. Algebraic Form

$$f_{XY}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\right.\right.\right.$$

Parameters

- μ_X, μ_Y: Means
- σ_X, σ_Y: Standard deviations
- ρ: Correlation coefficient

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3. Interpretation (Page 53–54) L18

Example:

- X: CO_2 concentration
- Y: Temperature

If:

$$\rho > 0 \Rightarrow \text{Higher } CO_2 \rightarrow \text{Higher temperature}$$

Applications

- Signal processing
- Communication systems
- Noisy measurements

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FINAL SUMMARY (MEMORY-OPTIMIZED)

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1. Key Concepts (Flashcard Style)

- Marginal PMF → "Sum out the other variable"
- Correlation → "Raw product measure"
- Covariance → "Centered dependence"
- Pearson → "Normalized covariance"
- Gaussian → "Bell-shaped joint behavior"

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2. Formulas to Remember

- $p_X(x) = \sum_y p_{XY}(x, y)$
- $E[XY]$
- $\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$
- $\rho = \frac{\text{Cov}}{\sigma_X \sigma_Y}$

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3. What to Learn

- Difference: correlation vs covariance
- Why centering is important
- Independence vs zero covariance
- Reading joint distributions

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4. Applications (Exam-Oriented)

- Marginalization → tables & joint distributions
- Covariance → detect direction of dependence
- Pearson → measure strength
- Gaussian → modeling real-world noise

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5. Tricks

- Independence ⇒ directly use $E[XY] = E[X]E[Y]$
- Linear transformation → use covariance property
- For tables → sum rows/columns quickly

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6. Be Careful:

- Zero covariance ≠ independence
- Correlation depends on scaling
- Forgetting to subtract mean in covariance
- Misusing joint vs marginal distributions
- Integration limits in continuous cases

END OF SCRIBE

Sources